ADAPTIVE SOFTWARE ENGINEERING (ASE) COURSE PROJECT SUBMISSION

PROJECT TITLE:

Context-Aware Self-Healing Distributed Systems Using Generative Graph Neural Networks for Dynamic Fault Remediation

Course Coordinator: G.Lavanya

TEAM DETAILS & ROLE ALLOCATION:

Member 1 (AI Architect):

• Name: Ch. Aditya

• Roll Number: 2420030669

Member 2 (Telemetry & Knowledge Base Lead):

• Name: A. Aaryan

• Roll Number / Student ID: 2420090098

Member 3 (Control Loop & Actuation Engineer):

• Name: K. Gokul

• Roll Number: 2420030716

Member 4 (Chaos Engineering & Benchmark Lead):

• Name: Nishanth

• Roll Number: 2420030679

==

PROJECT ABSTRACT

==

TITLE:

Context-Aware Self-Healing Distributed Systems Using Generative Graph Neural Networks for Dynamic Fault Remediation

ABSTRACT:

Modern cloud-native distributed microservice architectures face escalating operational complexity, where cascading component failures often trigger prolonged downtime and severe service degradation. This project presents an autonomous, context-aware self-healing framework rooted in Adaptive Software Engineering (ASE) principles to achieve end-to-end fault detection, root-cause localization, and dynamic remediation.

The architecture operationalizes a real-time MAPE-K (Monitor, Analyze, Plan, Execute, Knowledge) control loop integrated with Generative Graph Neural Networks (GNNs). During the monitoring and analysis stages, multi-dimensional telemetry—including CPU utilization, memory saturation, request latency, and inter-service HTTP/gRPC call graphs—is ingested at 100ms intervals across distributed nodes and transformed into dynamic spatiotemporal graph embeddings. The Generative GNN models complex topological dependencies to predict impending failure propagation, achieving a >96.5% target accuracy and an F1-score of 0.94 in root-cause anomaly detection, significantly outperforming conventional static threshold monitors.

In the planning and execution stages, the framework dynamically synthesizes context-aware remediation graphs. It executes targeted self-healing policies, such as automated container rescheduling, localized circuit breaking, and intelligent traffic rerouting, within a sub-500ms execution latency.

To validate system resilience, extensive chaos engineering scenarios were injected, encompassing container crashes, memory leaks, and network partition faults. Benchmarking demonstrates an 82% reduction in Mean Time to Resolution (MTTR), decreasing recovery intervals from several minutes to under 1.2 seconds, while eliminating over 85% of manual operational interventions. By defining strict Service Level Agreements (SLAs), the architecture guarantees predefined fault-resolution windows, ensuring that performance standards are measurable and explicitly quantified to sustain 99.99% high availability even under 30% concurrent node failure states.